leaf-conductance

Boundary-layer conductance of inclined elliptical leaves under free, forced and mixed convection

Model description, verification and validation

Software version 1.1.0 (Python and MATLAB)

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Abstract

`leaf-conductance` computes the boundary-layer conductances of a flat elliptical leaf for heat, water vapour and CO_2, separately for the upper and the lower leaf side. Its core is the free-convection model of Graefe et al. (2022), which accounts for leaf size, aspect ratio and a two-axis inclination through flux-weighted characteristic lengths. The software extends this core by (i) a closed-form solution of the characteristic-length integral for elliptical leaves, (ii) laminar forced convection along the wind streamlines projected onto the leaf plane (flowline concept), (iii) the laminar–transitional–turbulent flat-plate correlation of Lienhard (2020) with a turbulence-dependent transition onset (Mayle, 1991), (iv) a five-parameter angle-of-attack model for the windward and leeward leaf side fitted to 815 literature measurements, (v) a turbulent branch of free convection, (vi) mixed convection by power-mean blending and (vii) mode-dependent conversion of the heat conductance to scalar conductances. This document states every equation as implemented, the conventions of the two implementations, the test cases that verify them and the validation evidence, and it identifies the limits of validity. The Python and MATLAB implementations pass all 17 + 17 unit tests and agree on 573 shared reference cases to within 1×10^{-15} relative deviation.

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1 Introduction and scope

The leaf boundary-layer conductance g_b controls the exchange of sensible heat, water vapour and CO_2 between a leaf and the surrounding air and is therefore a central parameter of leaf energy-balance, transpiration and photosynthesis models. Simple formulations such as $g_b = |\Delta T|^{0.33}/400$ (Monteith, 1973) or the laminar flat-plate law $g_b \propto \sqrt{u/d}$ (Campbell and Norman, 1998) ignore the leaf shape and orientation and do not distinguish the two leaf sides. Graefe et al. (2022) derived a free-convection model for inclined elliptical leaves from computational fluid dynamics, which resolves both effects.

The repository `leaf-conductance` provides this model in Python (single module `graefe_convection.py`, NumPy only) and MATLAB (one function per file) and extends it to forced and mixed convection. Table 1 identifies which parts are taken from the peer-reviewed paper and which are extensions introduced in the software; the extensions have not been peer reviewed. Equation numbers of the paper are cited as "G22 Eq. x".

Table 1: Origin of the model components.

Component	Source	Section
Geometry, characteristic lengths L_h, L_i	Graefe et al. (2022), Eqs. 4–11, App. A/B	3
Closed form of the flux-weighted length	extension (exact solution of G22 Eq. 4)	3.4
Laminar free convection, both sides	Graefe et al. (2022), Eqs. 1, 3a–3g, 14, 18, Table 3	4
Turbulent free convection	extension	4.2
Forced convection, flowline length	extension	5
Laminar–turbulent transition	Lienhard (2020), Mayle (1991), Kottke et al. (1977)	5.3
Angle of attack, finite-plate factor	extension, fitted to literature data	5.4
Mixed convection	extension (cubic power mean)	6
Heat, H_2O and CO_2 conductances	generalisation of G22 Eq. 17; Massman (1998)	7

Notation and units. All quantities are in SI units; temperatures in K, lengths in m, angles in degrees at the user interface. Table 2 lists the symbols. The subscripts U and L denote the upper (adaxial in the horizontal position) and lower leaf side. Conductances and fluxes refer to one-sided leaf area.

Table 2: Principal symbols.

Symbol	Meaning	Unit	Symbol	Meaning	Unit
L, W	leaf length, width ($L \geq W$)	m	T_L, T_A	leaf, air temperature	K
S	one-sided area	m^2	ΔT	$T_L - T_A$ (or Eq. 17)	K
AR	aspect ratio L/W	–	u	wind speed	$m\,s^{-1}$
α_x, α_y	rotations about short / long axis	deg	TI	turbulence intensity	%
θ	angle leaf normal – vertical	rad	ν	kinematic viscosity	$m^2\,s^{-1}$
φ	streamline angle to short axis	rad	κ	thermal diffusivity	$m^2\,s^{-1}$
L_h, L_i	char. lengths, horizontal / inclined	m	ρ, c_p	air density, heat capacity	–
L_{Co}	mean chord length	m	Nu_F	flux-proportional Nusselt no.	m^{-1}
L_f	forced-convection streamline length	m	Ra	Rayleigh number per m^3	m^{-3}
β	angle of attack	deg	g_b, g_v, g_c	conductance heat, H_2O, CO_2	$m\,s^{-1}$

2 Physical basis

2.1 Conductance and flux-proportional Nusselt number

Following Graefe et al. (2022), the Nusselt number is expressed without a length scale, $Nu_F = h/k$ (m^{-1}), where h is the heat-transfer coefficient and k the thermal conductivity of air. The conductance for heat and the sensible heat flux of one leaf side are then (G22 Eq. 1, G22 Eq. 14)

$$g_b = \kappa\, Nu_F, \qquad H = \rho\, c_p\, g_b\, (T_L - T_A), \tag{1}$$

with $H > 0$ from leaf to air. A conventional dimensionless Nusselt number based on a length ℓ is $Nu_\ell = Nu_F\, \ell$. This formulation allows terms with different characteristic lengths to be combined.

2.2 Air properties

All air properties are evaluated at the film temperature $T_F = (T_L + T_A)/2$ for dry air (`air_properties`, `airProperties.m`):

$$\rho = \frac{p}{R_d T_F}, \quad R_d = 287.05\, J\, kg^{-1}\, K^{-1}, \tag{2}$$

$$\mu = \frac{1.458 \times 10^{-6}\, T_F^{1.5}}{T_F + 110.4} \text{ (Sutherland's law; Sutherland, 1893)}, \qquad \nu = \mu/\rho, \tag{3}$$

$$k = 0.02624\, (T_F/300)^{0.8646}, \qquad \kappa = k/(\rho\, c_p), \quad c_p = 1004\, J\, kg^{-1}\, K^{-1}, \tag{4}$$

with $Pr = 0.71$ and $g = 9.81\, m\, s^{-2}$ held constant. At 20 °C and 101325 Pa these give $\rho = 1.204\,\mathrm{kg\,m^{-3}}$, $\nu = 1.51 \times 10^{-5}$ and $\kappa = 2.12 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$ (test `test_air_properties_20C`).

3 Leaf geometry and characteristic lengths

3.1 Ellipse, dissection index and horizontal leaf

The leaf is a flat ellipse with axes $L \geq W$ (`leaf_geometry`, `leafGeometry.m`); $L < W$ raises an error. Its area is $S = \frac{\pi}{4} LW$ and its aspect ratio $AR = L/W$. The perimeter is expressed through the dissection index (G22 Eq. 8, maximum error 0.04 % for $AR = 1$–4)

$$DI = 1.112 \left(\frac{L + W}{\sqrt{LW}} \right)^{0.783} - 0.913, \qquad P = 2\, DI \sqrt{\pi S}, \tag{5}$$

and the characteristic length of the horizontal leaf is (G22 Eq. 7)

$$L_h = \frac{S}{P} = \frac{\sqrt{S}}{2\, DI \sqrt{\pi}}. \tag{6}$$

For a circle $DI = 1$ and $L_h = D/4$.

3.2 Inclination and streamline direction

The unrotated leaf has its short axis along x, its long axis along y and its normal along z (up). It is first rotated about y by α_y and then about x by α_x, $\mathbf{R} = \mathbf{R}_x(\alpha_x)\, \mathbf{R}_y(\alpha_y)$ (App. B of Graefe et al., 2022). The rotated normal $\mathbf{n} = \mathbf{R}\hat{\mathbf{z}}$ subtends the angle

$$\cos\theta = \cos\alpha_x \cos\alpha_y \qquad \text{(G22 Eq. 11)} \tag{7}$$

with the vertical. Buoyancy-driven streamlines follow the steepest-ascent direction $\mathbf{s} = \hat{\mathbf{z}} - (\hat{\mathbf{z}}\cdot\mathbf{n})\mathbf{n}$ in the leaf plane; its angle to the rotated short axis $\mathbf{w} = \mathbf{R}\hat{\mathbf{x}}$ is

$$\varphi = \arccos \frac{|\mathbf{s}\cdot\mathbf{w}|}{\|\mathbf{s}\|} \in [0, \pi/2]. \tag{8}$$

For $\alpha_y = 0$ the flow runs along the long axis ($\varphi = \pi/2$). For a horizontal leaf $\mathbf{s} = \mathbf{0}$ and $\varphi = \pi/2$ is returned by convention (the inclined term vanishes with $\sin\theta = 0$).

3.3 Mean chord length and characteristic length of the inclined leaf

The mean length of the chords parallel to the streamline direction is (G22 Eq. 9, App. A; written with sin / cos to remain finite at $\varphi = \pi/2$)

$$L_{Co}(\varphi) = \frac{\pi LW}{4\sqrt{(W\sin\varphi)^2 + (L\cos\varphi)^2}}, \tag{9}$$

and the characteristic length of the inclined leaf is the linear fit (G22 Eq. 9)

$$L_i = (0.04511\, n + 1.006)\, L_{Co}(\varphi), \qquad n = 3 \times 0.25 = 0.75. \tag{10}$$

For $\alpha_y = 0$, $L_i = 0.816\, L$. Figure 2a shows L_{Co} over φ.

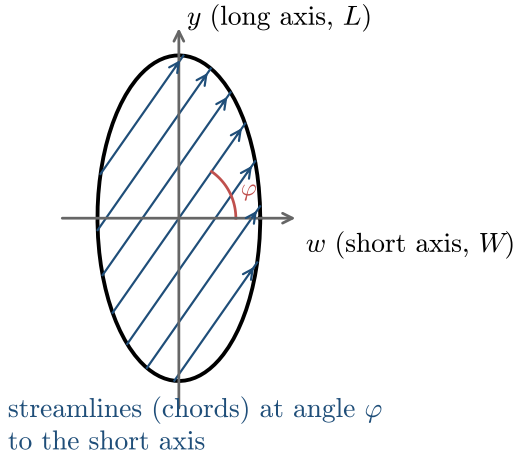


Figure 1: Leaf in its own plane. Streamlines (chords) at the angle φ to the short axis; along the chord at offset t the boundary layer develops over the length $D(t)$. Free convection uses the steepest-ascent direction, forced convection the wind direction projected onto the leaf plane.

3.4 Closed form of the flux-weighted characteristic length

G22 Eq. 4 defines the characteristic length as the length for which a uniform transfer law $h \propto \ell^{n-1}$ (local Nusselt number $\propto \ell^n$) yields the same total flux as the integral over all chords. For an ellipse the chords parallel to any direction have the lengths $D(t) = D_0\sqrt{1-(t/t_{\max})^2}$, with $D_0 = \frac{4}{\pi}L_{Co}(\varphi)$ the longest chord. Substituting $t = t_{\max}\sin\psi$ gives

$$\frac{\int D(t)^n\,\mathrm{d}t}{\int D(t)\,\mathrm{d}t} = D_0^{n-1}\,\frac{\int_0^{\pi/2}\cos^{n+1}\psi\,\mathrm{d}\psi}{\int_0^{\pi/2}\cos^2\psi\,\mathrm{d}\psi} \;\Rightarrow\; L_c^{n-1} = D_0^{n-1}\,\frac{2}{\pi}\,B\!\left(\tfrac{1}{2}, \tfrac{n}{2}+1\right), \tag{11}$$

with the Beta function B (`characteristic_length`, `characteristicLength.m`; for $n = 1$ the mean chord length is returned). Because the chord distribution of an ellipse is self-similar, L_c/L_{Co} depends only on n and not on φ or AR:

- $n = 0.75$ (laminar free convection, $Nu \propto Ra^{1/4}$): $L_c/L_{Co} = 1.0395$; the fit of Eq. (10) gives 1.0398, i.e. the paper's regression is reproduced within 0.03 % for all φ and AR (Fig. 2b).
- $n = 0.5$ (laminar forced convection, $h \propto \ell^{-1/2}$): $L_f/L_{Co} = 1.0281$.

The model keeps the published fit (Eq. 10) for free convection and uses Eq. (11) with $n = 0.5$ for forced convection.

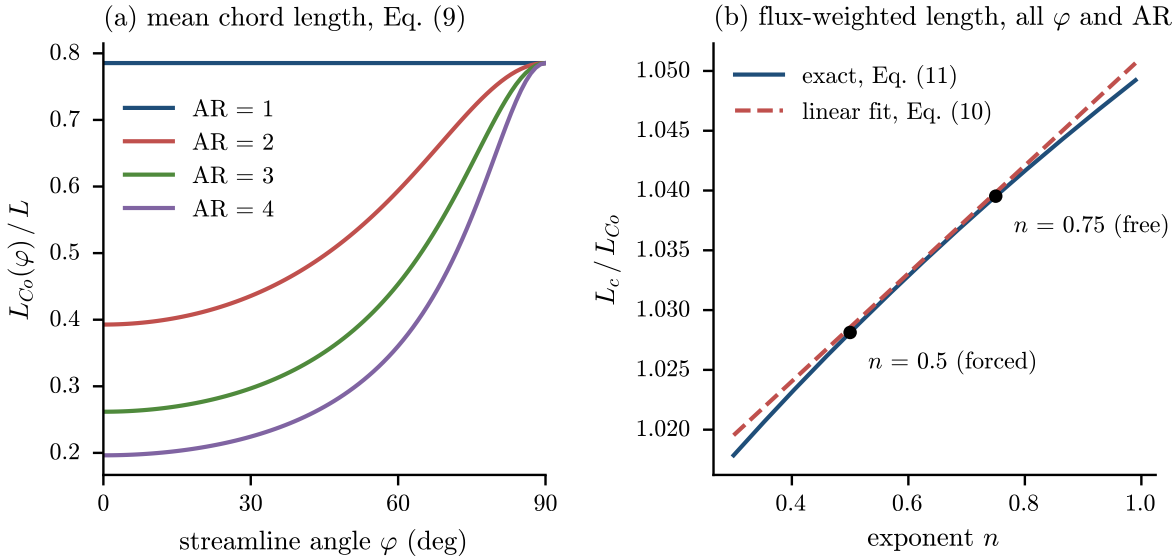


Figure 2: (a) Mean chord length L_{Co} relative to the leaf length for aspect ratios 1–4. (b) Ratio of the flux-weighted characteristic length to the mean chord length over the exponent n: closed form Eq. (11) and the linear fit Eq. (10). The ratio is the same for all streamline angles and aspect ratios; at $n = 0.75$ fit and closed form differ by 0.03 %.

4 Free convection

4.1 Laminar model of Graefe et al. (2022)

With the Rayleigh number per unit volume (G22 Eq. 3g; buoyancy coefficient $1/T_A$, ν at film temperature)

$$Ra = \frac{g\,|\Delta T|\,Pr}{T_A\,\nu^2} \quad [\mathrm{m}^{-3}], \tag{12}$$

the contributions of the inclined (i) and horizontal (h) components for a *heated* leaf ($\Delta T > 0$) are (G22 Eq. 3b–G22 Eq. 3e)

$$Nu_{i,U} = p_3\,L_i^{-1/4}\,\sin^{p_4}\theta\;Ra^{1/4}, \qquad Nu_{i,L} = p_3\,L_i^{-1/4}\,\sin^{1/4}\theta\;Ra^{1/4}, \tag{13}$$

$$Nu_{h,U} = p_5\,L_h^{-1/4}\,\cos^{1/4}\theta\;Ra^{1/4}, \qquad Nu_{h,L} = p_6\,L_h^{-0.4}\,\cos^{0.2}\theta\;Ra^{0.2}, \tag{14}$$

the conduction limit of the elliptical plate (G22 Eq. 3f; Yovanovich, 1998) is

$$Nu_c = \frac{5.3\,\sin(0.0338\,AR + 0.607)}{\sqrt{S}}, \tag{15}$$

and the side-specific flux-proportional Nusselt number is the nested power mean (G22 Eq. 3a)

$$Nu_F = \left\{\left[Nu_i^{p_1} + Nu_h^{p_1}\right]^{p_2/p_1} + Nu_c^{p_2}\right\}^{1/p_2}. \tag{16}$$

For a *cooled* leaf ($\Delta T < 0$) the right-hand sides of U and L are exchanged, since the lower side then faces the unstable stratification. The parameters (Table 3 of the paper) are $p_1 = 6$, $p_2 = 1.42$, $p_3 = 0.513$, $p_4 = 0.4$, $p_5 = 0.202$, $p_6 = 0.492$. Implementation: `free_convection`, `nusselt_free` / `graefeFreeConvection.m`. To avoid a singularity at $\Delta T = 0$, $|\Delta T|$ is bounded below by 10^{-9} K.

Vapour buoyancy. If leaf-surface and air vapour pressures e_L, e_A are supplied, the buoyancy uses the virtual temperature difference (G22 Eq. 18)

$$\Delta T' = \Delta T + 0.38\,\frac{e_L T_L - e_A T_A}{p}, \tag{17}$$

so that a transpiring leaf slightly cooler than the air may still drive upward convection. The heat flux in Eq. (1) always uses the actual $T_L - T_A$.

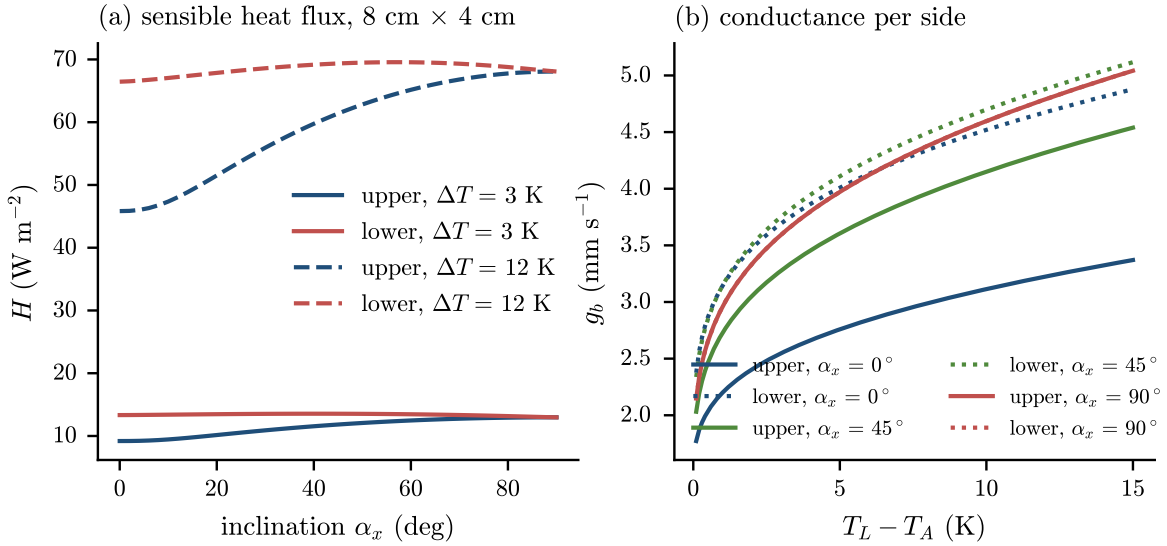


Figure 3: Laminar free convection of an 8×4 cm leaf at $T_A = 20\,°C$. (a) Sensible heat flux of both sides over the inclination α_x ($\alpha_y = 0$), cf. Fig. 6b of Graefe et al. (2022): the sides converge with increasing inclination and are identical for the vertical leaf. (b) Conductance per side over the temperature difference.

4.2 Extension: turbulent free convection

For large or hot leaves the buoyant boundary layer on the unstable side becomes turbulent, where the heat transfer becomes independent of the length scale. `free_convection_transition` / `freeConvectionTransition.m` add

$$Nu_{F,\mathrm{turb}} = 0.14\,Ra^{1/3}, \qquad Nu_F = \left(Nu_{F,\mathrm{lam}}^6 + Nu_{F,\mathrm{turb}}^6\right)^{1/6} \tag{18}$$

on the buoyantly unstable side (upper side of a heated, lower side of a cooled leaf, both sides of a vertical leaf). The stably stratified side keeps its laminar value. The turbulent share $s_t = Nu_{F,\mathrm{turb}}^6/(Nu_{F,\mathrm{lam}}^6 + Nu_{F,\mathrm{turb}}^6)$ is reported for the scalar exponent (Section 7). For leaf-sized objects s_t is negligible (e.g. $< 2\,\%$ change for an 8×4 cm leaf at $\Delta T = 1$ K); the branch matters for leaves of decimetre size and $\Delta T \gtrsim 10$ K.

5 Forced convection

5.1 Wind geometry and flowline length

The horizontal wind blowing from the meteorological direction δ (`wind_from_deg`) has the direction vector $\mathbf{d}$ towards azimuth $\delta + 180°$. The leaf azimuth A (`leaf_azimuth_deg`) is the compass direction that the upper surface faces (180 = south); before tilting, the long axis points horizontally towards $A + 180°$ (up-slope) and the short axis is perpendicular to it. Expressed in the leaf frame, $\mathbf{d}$ is projected onto the leaf plane,

$$\mathbf{d}_p = \mathbf{d} - (\mathbf{d}\cdot\mathbf{n})\,\mathbf{n}, \qquad \varphi_w = \arccos\frac{|\mathbf{d}_p\cdot\mathbf{w}|}{\|\mathbf{d}_p\|}, \qquad \beta = \arcsin|\mathbf{d}\cdot\mathbf{n}|, \tag{19}$$

where φ_w is the streamline angle to the short axis (`wind_streamline_angle`), $\|\mathbf{d}_p\|$ the in-plane fraction of the wind and β the angle of attack (`wind_attack_angle`); the windward side is U if $\mathbf{d}\cdot\mathbf{n} < 0$, otherwise L. The streamline length for laminar forced convection is Eq. (11) with $n = 0.5$,

$$L_f = L_c(\varphi_w;\, n = 0.5) \qquad (\texttt{forced_length}). \tag{20}$$

If the wind direction is unknown (`wind_from_deg=None` / `[]`) or the wind is normal to the leaf ($\|\mathbf{d}_p\| < 10^{-6}$), the orientation-averaged chord length $\pi S/P$ (G22 Eq. 5) replaces L_{Co}, i.e. $D_0 = 4S/P$. The full wind speed u is used in the Reynolds number; the in-plane fraction is reported but not applied.

5.2 Laminar flat plate

In the basic forced model (`nusselt_forced` with `transition=False`) each side follows the laminar flat-plate law (Pohlhausen, 1921)

$$Nu_{F,\text{forced}} = \frac{c\,Re^{1/2}\,Pr^{1/3}}{L_f}, \qquad Re = \frac{uL_f}{\nu}, \qquad c = 0.664. \tag{21}$$

(Campbell and Norman (1998) use $c = 0.6$; the coefficient can be passed as `c_forced`.)

5.3 Laminar–turbulent transition

`forced_nusselt_lienhard` / `forcedNusseltLienhard.m` implement the average Nusselt number of Lienhard (2020). The local laws are

$$\text{laminar } (Re_x \le Re_l): \quad Nu_x = a_l\,Re_x^{1/2}Pr^{1/3}, \quad a_l = 0.332, \tag{22}$$

$$\text{transitional } (Re_l < Re_x \le Re_u): \quad Nu_x = Nu_x(Re_l)\,(Re_x/Re_l)^c, \quad c = 0.9922\log_{10}Re_l - 3.013, \tag{23}$$

$$\text{turbulent } (Re_x > Re_u): \quad Nu_x = 0.0296\,Re_x^{0.8}Pr^{0.6}. \tag{24}$$

The end of transition Re_u follows from continuity of Nu_x at Re_u,

$$Re_u = \left[\frac{a_l}{0.0296}\,Re_l^{1/2-c}\,Pr^{1/3-0.6}\right]^{1/(0.8-c)}, \tag{25}$$

and the plate average $\overline{Nu} = \int_0^{Re} Nu_x\,Re_x^{-1}\,\mathrm{d}Re_x$ is evaluated exactly:

$$\begin{aligned}\overline{Nu} = {} & 2a_l Pr^{1/3}\min(Re, Re_l)^{1/2} + \frac{Nu_x(Re_l)}{c}\left[\left(\frac{\min(Re, Re_u)}{Re_l}\right)^c - 1\right]_{Re>Re_l} \\ & + 0.037\,Pr^{0.6}\left[Re^{0.8} - Re_u^{0.8}\right]_{Re>Re_u}.\end{aligned} \tag{26}$$

For a uniform-heat-flux plate the laminar average coefficient becomes 0.680 instead of 0.664 (`bc="uniform_flux"`); an unheated starting length x_0 multiplies the laminar part by $[1 - (x_0/L)^{3/4}]^{-1/3}$.

Transition onset. The onset depends on the free-stream turbulence intensity. Mayle's (1991) correlation $Re_{\theta,t} = 400\,\mathrm{TI}^{-5/8}$, converted with the Blasius momentum thickness $Re_\theta = 0.664\,Re_x^{1/2}$, gives

$$Re_l = 3.63\times10^5\,\mathrm{TI}^{-1.25}, \qquad 2\times10^3 \le Re_l \le 5\times10^5 \qquad (\texttt{transition_onset_reynolds}). \quad (27)$$

Example values: $\mathrm{TI} = 0.1\,\% \to 5\times10^5$(cap), $1\,\% \to 3.63\times10^5$, $5\,\% \to 4.86\times10^4$, $10\,\% \to 2.04\times10^4$, $20\,\% \to 8.58\times10^3$, $50\,\% \to 2.73\times10^3$. In canopies (TI 20–50 %) leaves of a few centimetres can therefore reach transitional flow at moderate wind speeds. The regime is reported as `forced_regime` (laminar / transitional / turbulent).

Blunt leading edges. If $Re_t = u\,t/\nu \ge 2000$ for a plate of thickness t (Kottke et al., 1977), the boundary layer is tripped at the edge and `forced_nusselt_turbulent` gives $\overline{Nu} = 0.037\,Re^{0.8}\,Pr^{0.6}$. Thin leaves do not reach this criterion; it is provided for thick test bodies only and is not used in the general model.

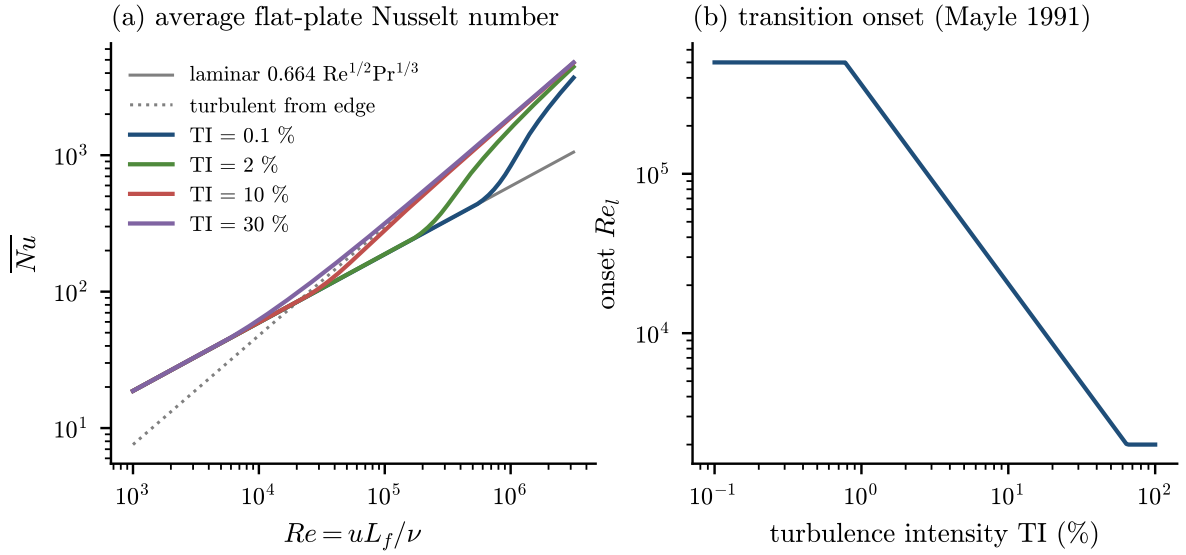


Figure 4: (a) Average flat-plate Nusselt number (Eq. 26) for several turbulence intensities, with the laminar and fully turbulent limits. (b) Transition onset Reynolds number after Mayle (1991), bounded to $[2\times10^3, 5\times10^5]$.

5.4 Extension: angle of attack and finite-plate enhancement

Inclined to the wind, the windward side of a thin plate experiences an accelerated outer flow, while the leeward side is first covered by a reattaching separation bubble (enhancement at small angles) and at larger angles by a wake (reduction). A parsimonious model with five parameters describes both sides (`angle_of_attack_factors`, `forced_nusselt_leaf` /

`angleOfAttackFactors.m`, `forcedNusseltLeaf.m`):

$$\overline{Nu}_0 = f_e\,\overline{Nu}_{\text{plate}}(Re, \text{TI}), \tag{28}$$

$$\text{windward:}\quad \overline{Nu}_W = \overline{Nu}_0\, g_W, \qquad g_W = 1 + a_W \sin\beta, \tag{29}$$

$$\text{leeward:}\quad \overline{Nu}_B = \overline{Nu}_0\, g_B, \qquad g_B = w\,(1 + a_R \sin 2\beta) + (1 - w)\, c_w, \tag{30}$$

$$w = \exp\!\left[-(\beta/\beta_s)^2\right], \tag{31}$$

where $\overline{Nu}_{\text{plate}}$ is Eq. (26) and f_e accounts for edge effects and three-dimensional flow on finite plates. The fitted values are given in Table 8. In parallel flow both factors are 1; the leeward factor peaks at $g_B = 1.196$ at $\beta = 17.0°$ and tends to $c_w < 1$ in normal flow (Fig. 5).

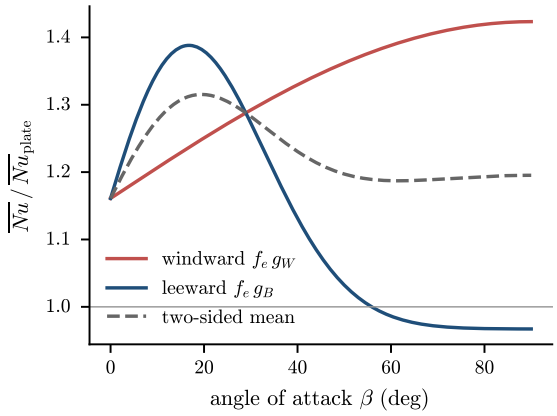


Figure 5: Angle-of-attack model: average Nusselt number of the windward and leeward side relative to the infinite flat plate in parallel flow, including the finite-plate factor $f_e = 1.161$.

6 Mixed convection

Free and forced contributions are combined per side by the cubic power mean

$$Nu_{F,\text{mix}} = \left(Nu_{F,\text{free}}^3 + Nu_{F,\text{forced}}^3\right)^{1/3}, \qquad s_f = \frac{Nu_{F,\text{forced}}^3}{Nu_{F,\text{free}}^3 + Nu_{F,\text{forced}}^3}, \tag{32}$$

where s_f is the forced share (`mixed_convection` / `graefeMixedConvection.m`). Assisting and opposing flows are not distinguished. Options select the transition model (`transition`, `TI_pct`), the angle-of-attack model (`angle_model`; the windward side receives g_W, the other side g_B) and the turbulent free-convection branch (`free_transition`). Without wind ($u = 0$) the result is identical to pure free convection.

7 Conductances for heat, water vapour and CO_2

For a transfer law $Nu \propto Pr^m$ (or Sc^m), the conductance for a scalar with molecular diffusivity D_s is

$$g_s = g_b\,(D_s/\kappa)^k, \qquad k = 1 - m. \tag{33}$$

The exponent differs between transport modes (Table 3). The general model (`leaf_conductances` / `leafConductances.m`) uses the share-weighted mean, generalising G22 Eq. 17:

$$s_c = Nu_c / \max(Nu_{F,\text{lam}}, Nu_c), \qquad k_{\text{free,lam}} = 0.75 + 0.25\, s_c, \tag{34}$$

$$k_{\text{free}} = (1 - s_t)\, k_{\text{free,lam}} + s_t \cdot \tfrac{2}{3}, \qquad k_{\text{forced}} = (1 - f_t) \cdot \tfrac{2}{3} + f_t \cdot 0.4, \tag{35}$$

$$k = (1 - s_f)\, k_{\text{free}} + s_f\, k_{\text{forced}}, \tag{36}$$

with the conduction share s_c, the turbulent free share s_t (Section 4.2), the turbulent forced share f_t = turbulent part / total of Eq. (26), and the forced share s_f. The diffusivities (Massman, 1998) are

$$D_{\mathrm{H_2O}} = 2.12 \times 10^{-5}\, f, \qquad D_{\mathrm{CO_2}} = 1.39 \times 10^{-5}\, f \quad [\mathrm{m^2\,s^{-1}}], \qquad f = \left(\frac{T_F}{273.15}\right)^{1.75} \frac{101325}{p}. \tag{37}$$

Since $D_{\mathrm{H_2O}} > \kappa > D_{\mathrm{CO_2}}$, $g_v > g_b > g_c$ always.

Table 3: Diffusivity exponent k of the transport modes.

Mode	Law	k
Conduction limit	$Nu = \mathrm{const}$	1
Laminar free convection	$Nu \propto Ra^{1/4}$	3/4
Turbulent free convection	$Nu \propto Ra^{1/3}$	2/3
Forced, laminar and transitional	$Nu \propto Pr^{1/3}$	2/3
Forced, turbulent	$Nu \propto Pr^{0.6}$	0.4

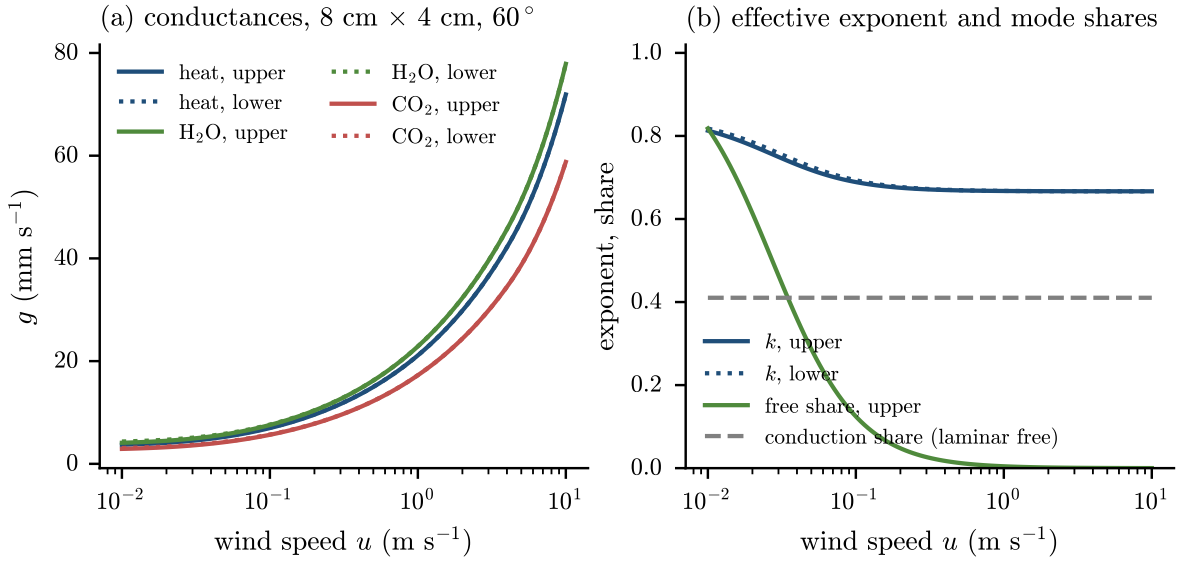


Figure 6: General model for an 8×4 cm leaf inclined by 60°, $\Delta T = 3$ K, $T_A = 20\,°C$, wind from the west (along the short axis, $\beta = 0$), TI = 20 %. (a) Conductances for heat, H_2O and CO_2 of both sides over the wind speed. (b) Effective exponent k, free share and conduction share.

8 Implementation

8.1 Structure and interfaces

Both implementations share the same formulation and parameter values; Table 4 maps the functions. Temperatures, vapour pressures and wind speed may be arrays (elementwise evaluation); the geometry is scalar. All functions return dictionaries (Python) or structs (MATLAB) containing the side-specific results and the intermediate quantities.

Table 4: Functions of the Python module `graefe_convection` and the MATLAB folder.

Python	MATLAB	Purpose
`air_properties`	`airProperties`	ρ, ν, κ, k
`leaf_geometry`	`leafGeometry`	S, AR, DI, P, L_h, θ, φ, L_{Co}, L_i
`characteristic_length`	`characteristicLength`	closed form Eq. (11)
`wind_streamline_angle`	`windStreamlineAngle`	φ_w, in-plane fraction
`wind_attack_angle`	`windAttackAngle`	β, windward side
`free_convection`	`graefeFreeConvection`	laminar free convection (paper)
`free_convection_transition`	`freeConvectionTransition`	with turbulent branch
`forced_nusselt_lienhard`	`forcedNusseltLienhard`	Eq. (26)
`angle_of_attack_factors`	`angleOfAttackFactors`	g_W, g_B
`forced_nusselt_leaf`	`forcedNusseltLeaf`	windward/leeward Nusselt numbers
`mixed_convection`	`graefeMixedConvection`	free + forced, per side
`leaf_conductances`	`leafConductances`	general model, heat/H_2O/CO_2

Defaults. `mixed_convection` defaults to the laminar forced law without angle model. `leaf_conductances` enables all extensions (`transition`, `free_transition`, `angle_model` = true) with TI = 1 % unless specified.

Usage. Python and MATLAB calls of the general model:

```python
from graefe_convection import leaf_conductances
r = leaf_conductances(TL=296.15, TA=293.15, u=1.5, L=0.08, W=0.04,
                      alpha_x_deg=60, leaf_azimuth_deg=180,
                      wind_from_deg=270, TI_pct=20)
r["gb_U"], r["gv_U"], r["gc_U"], r["k_U"]   # m/s, upper side
```

```
r = leafConductances(296.15, 293.15, 1.5, 0.08, 0.04, 60, 0, 180, 270, ...
                     [], [], [], 20);
```

8.2 Worked example

Table 5 documents intermediate and final values for an 8×4 cm leaf inclined by $\alpha_x = 60°$, $T_L = 23\,°C$, $T_A = 20\,°C$, $p = 101325\,Pa$, facing south, so that users can check other implementations.

Table 5: Worked example (8 cm × 4 cm, $\alpha_x = 60°$, $\alpha_y = 0$, $\Delta T = 3$ K, $T_A = 293.15$ K).

Geometry and air		Free convection (laminar, paper)	
S (cm^2)	25.133	Ra (m^{-3})	3.086×10^8
DI	1.091	Nu_c (m^{-1})	66.031
L_h (mm)	12.966	$Nu_{F,U}$ / $Nu_{F,L}$ (m^{-1})	160.930 / 173.664
L_{Co} / L_i (mm)	62.832 / 65.335	$g_{b,U}$ / $g_{b,L}$ (mm s^{-1})	3.457 / 3.730
θ / φ (deg)	60.000 / 90.000	H_U / H_L (W m^{-2})	12.473 / 13.460
ν, κ (10^{-5} m^2 s^{-1})	1.520, 2.148		
ρ (kg m^{-3})	1.198		
Mixed, laminar, $u = 1$ m s^{-1} from W		**General model, TI = 20 %, wind from W**	
φ_w (deg)	0.000	$u = 0$: g_b / g_v / g_c, U (mm s^{-1})	3.480 / 3.850 / 2.695
L_f (mm)	32.300	$u = 0$: k_U	0.845
Re	2125	$u = 3$: g_b / g_v / g_c, U (mm s^{-1})	36.530 / 39.560 / 29.855
$g_{b,U}$ / $g_{b,L}$ (mm s^{-1})	18.202 / 18.213	$u = 3$: k_U, Re, regime	0.667, 6376, laminar
forced share s_f	0.992		

Note that in the general model with $u = 0$ the upper-side heat conductance ($3.480\,\mathrm{mm\,s^{-1}}$) slightly exceeds the laminar value (3.457) because of the turbulent free-convection branch; the wind from the west runs along the short axis of the south-facing leaf ($\varphi_w = 0$, $\beta = 0$), so both sides receive the same forced contribution.

9 Verification

Verification establishes that the code implements the equations above correctly. It consists of (i) unit tests of analytical limits, symmetries and published values, run independently for each implementation, and (ii) a cross-implementation regression test on a shared set of reference cases.

9.1 Unit tests

The Python suite (`python/test_graefe_convection.py`, pytest) and the MATLAB suite (`matlab/tests/testGraefeFreeConvection.m`, `matlab.unittest`) contain 17 tests each, listed with their verified properties in Table 6. Tolerances are given in the source; analytic identities are checked to 10^{-9}–10^{-15}, published regression values to their stated precision.

Table 6: Unit tests (Python / MATLAB) and the properties they verify.

Python test	MATLAB test	Verified property
`geometry_circle`	`geometryCircle`	circle: $DI = 1 \pm 0.005$, $L_h = D/4$, $S = \pi r^2$
`geometry_ellipse_axes_and_li`	`geometryEllipse`	AR; $\varphi = 90°$ for $\alpha_y = 0$; $L_{Co} = \pi L/4$; $L_i = 0.816L$ (paper); θ from Eq. 11; $0 < \varphi < 90°$ and smaller L_i for $\alpha_y = 45°$; error for $L < W$
`air_properties_20C`	`airAt20C`	ρ, ν, κ, k at 20 °C against tabulated values
`vertical_plate_matches_p3`	`verticalPlate`	vertical leaf: both sides identical; $Nu_F L_i = p_3 Ra_{L_i}^{1/4}$ up to the conduction blending ($< 30\,\%$)
`cooled_leaf_swaps_sides`	`cooledLeafSwapsSides`	sign reversal of ΔT exchanges U and L; heated horizontal leaf: upper side weaker ($p_5 < p_6$)
`fig6_magnitudes_and_convergence`	`fig6Behaviour`	Fig. 6b of the paper: H in plausible range, monotone convergence of both sides with α_x, identical at 90°; superlinear increase of H with ΔT
`vectorised_and_eq18`	`vectorisedAndEq18`	array inputs; sign of H follows ΔT; Eq. 18 makes a transpiring, cooler leaf buoyant; Monteith reference
`exact_characteristic_length_matches_eq9`	`exactCharacteristicLength`	Eq. (11) vs. Eq. 9 ($< 0.3\,\%$) for 3 shapes × 3 angles; $L_i = 0.816L$; $L_f/L_{Co} = 1.0286 \pm 0.002$; $n = 1$ limit
`wind_streamline_angle`	`windStreamlines`	N/E/NE wind on a horizontal south-facing leaf gives $\varphi_w = 90/0/45°$; wind normal to a vertical leaf has no in-plane part; in-plane fraction $\cos 60°$; normal-flow fallback
`forced_and_mixed`	`mixedConvection`	laminar law identity; $u = 0$ reproduces free convection; strong wind: forced share > 0.9; cubic blending inequality; L_f along $>$ unknown $>$ across; monotone in u
`lienhard_transition`	`lienhardTransition`	onset bounds and value at TI = 5 %; laminar below onset; between laminar and fully turbulent above; continuity and monotonicity on 400 points; TI and uniform-flux effects; reported regimes; turbulent free branch: no effect at low Ra, $> 30\,\%$ at high Ra, stable side unchanged; MATLAB: three values against Python

Python test	MATLAB test	Verified property
`angle_of_attack_model`	`angleOfAttackModel`	factors 1 in parallel flow; $g_W(90°) = 1 + a_W$; leeward peak > 1.15 at 10–35°, < 1 at 90°; f_e scaling; β and windward side for horizontal, vertical and inclined leaves; windward side stronger at 60°
`scalar_conductances`	`scalarConductances`	D_{H_2O}/D_{CO_2}; pure-mode exponents 1, 3/4, 2/3, 0.4; $g_v > g_b > g_c$; $k \to 2/3$ in laminar wind; turbulent forced and free shares lower k; vectorisation
`reference_values`	`matchesPythonReference`	121 free-convection reference cases (rel. tol. 10^{-9})
`reference_values_mixed`	`matchesPython ReferenceMixed`	320 mixed-convection cases
`reference_values_angle`	`matchesPython ReferenceAngle`	60 transition + angle-of-attack cases
`reference_values_ scalar`	`matchesPython ReferenceScalar`	72 general-model cases (heat, H_2O, CO_2)

Results. On 13 October 2026 all 17 Python tests passed (Python 3.12.6, NumPy 2.2.6, pytest) and all 17 MATLAB tests passed (MATLAB 25.2.0.3042426 (R2025b) Update 1). Continuous integration (GitHub Actions, `.github/workflows/tests.yml`) runs the Python suite on Python 3.9 and 3.12 on every push and pull request; the MATLAB suite is run locally (`runTests`).

9.2 Cross-implementation reference cases

`python/make_reference.py` evaluates the Python model on a factorial design and stores 573 cases in `reference/reference_values.json` (Table 7). Both test suites recompute every case and compare all stored outputs with the criterion $|a - b|/\max(1, |b|) \leq 10^{-9}$. Table 7 lists the largest deviation found; MATLAB and Python agree to the level of floating-point round-off.

Table 7: Reference case sets and maximum deviation $|a - b|/\max(1, |b|)$ from the stored values.

Set	Cases	Design	Values	Python	MATLAB
free	121	3 shapes (8×4, 8.4×8.4, 12×4 cm) $\times$ $\alpha_x \in \{0, 30, 60, 90\} \times \alpha_y \in \{0, 45\} \times$ $\Delta T \in \{-12, -3, 1, 3, 12\}$ K, + 1 case with Eq. 18	968	1.5×10^{-16}	8.2×10^{-16}
mixed	320	2 shapes $\times$ 4 orientations $\times$ 2 azimuths $\times$ 5 wind directions (incl. unknown) $\times$ $u \in \{0.3, 2\}$ $\times$ $\Delta T = \pm 3$ K	2880	4.8×10^{-16}	4.8×10^{-16}
angle	60	4 orientations $\times$ 5 wind directions $\times$ (u, TI) $\in \{(0.5, 1), (3, 10), (12, 30)\}$; 40 laminar, 20 turbulent	480	5.3×10^{-16}	5.7×10^{-16}
scalar	72	$\Delta T \in \{-6, 3, 12\} \times 4$ (u, TI) $\times$ 3 orientations $\times$ 2 sizes (8×4, 60×30 cm)	864	1.4×10^{-17}	2.2×10^{-16}

The Python column reflects the regression of the current Python environment against the stored values; deviations at the level of round-off arise from floating-point differences between library versions. It guards against unintended changes of the model. Because the reference values are generated by the Python code itself, this test verifies the consistency of the two implementations and the stability of the results, not the correctness of the formulation; the latter is addressed by the unit tests and the validation below.

10 Validation

10.1 Free-convection core

The laminar free-convection model and its parameters (Table 3) were derived and validated by Graefe et al. (2022) against CFD simulations of elliptical leaves and against the measurements of Hassani and Hollands (1989). The present implementation reproduces the qualitative and quantitative behaviour of Fig. 6b of the paper (Fig. 3a, unit test `fig6_magnitudes_and_convergence`) and the published value $L_i = 0.816L$.

As an additional check, the laminar model was compared with free-convection measurements of horizontal disks and squares (Ra 18 to 7.6×10^7; Hassani and Hollands, 1989) and of inclined 2 cm disks (Kobus and Wedekind, 2002); horizontal disks of 5–20 mm (Kobus and Wedekind, 2001) were used as an additional check. The model underpredicts the data of Hassani and Hollands (1989) by about 20 % and those of Kobus and Wedekind (2002) by about 10 %, which is consistent with the model assigning no transfer to the leaf edges, whereas the test bodies have heated edges. With the turbulent $Ra^{1/3}$ branch the deviation from Hassani and Hollands (1989) is reduced to −14 %.

10.2 Forced-convection extensions

Data. The transition, angle-of-attack and finite-plate parameters were fitted on 12 October 2026 to 1169 digitised literature measurements of mean Nusselt numbers of finite flat plates, disks and leaf models in forced convection (Chen et al., 1988a,b; Haseba, 1973; Haseba and Ito, 1984; Sparrow and Tien, 1977; Sparrow et al., 1979; Parkhurst et al., 1968; Powell, 1940; Schuepp, 1993; Ramirez et al., 2002; Kobus and Shumway, 2006; Motwani et al., 1985; aus der Wiesche and Helcig, 2016). The data cover $Re = 1.3 \times 10^3$–1.6×10^5, angles of attack 0–90°, windward, leeward and two-sided measurements and turbulence intensities 0.2–22 %. Heat and mass transfer measurements (naphthalene sublimation, evaporation, electrochemical method) were converted to Nu at $Pr = 0.71$ by the heat–mass transfer analogy. The fit set comprises 815 rows of thin plates with $Re\,t/L < 2000$ and wind-tunnel blockage ≤ 0.12; the thick bodies of aus der Wiesche and Helcig (2016) and Motwani et al. (1985) are excluded as outside the model's scope.

Procedure. The windward/leeward factors are applied to $f_e\,\overline{Nu}_{\mathrm{plate}}$ (Eq. 26, with the reported TI); for two-sided measurements the mean of both sides is used and the conduction limit is blended in (Yovanovich and Jafarpur, 1993). The five parameters minimise $\sum[\ln(Nu_{\mathrm{pred}}/Nu_{\mathrm{obs}})]^2$.

Table 8: Fitted parameters of the angle-of-attack model (815 measurements).

Parameter	Value	Std. error	Meaning
f_e	1.161	0.016	finite-plate enhancement (edges, 3-D flow)
a_W	0.226	0.029	windward enhancement by the accelerated outer flow
a_R	0.574	0.095	leeward enhancement by the reattaching separation bubble
β_s	31.2°	2.6°	transition from reattaching to wake regime
c_w	0.833	0.021	wake-regime level relative to parallel flow

Goodness of fit. The relative root-mean-square error is 23.9 % with a bias of −3.8 % and a standard deviation of the log ratio of 0.19. For comparison, the laminar flat plate without enhancement and angle terms gives 37.3 % and −21.9 %. In parallel flow alone (143 rows) the

model has a bias of −1.9 % and an rRMSE of 11.4 % (without f_e: −15.3 % and 20.2 %), which confirms that finite plates transfer about 16 % more heat than the infinite-plate law predicts. Tables 9 and 10 show the results by source and by angle of attack; no systematic trend with the angle remains.

Table 9: Validation statistics by source (rRMSE and bias of predicted vs. observed Nu). Face: b = both sides (two-sided measurement), w = windward, l = leeward.

Source	Face	n	rRMSE (%)	Bias (%)
Chen et al. (1988a)	b	42	7.8	−2.7
Chen et al. (1988b)	b / w / l	99 / 39 / 33	16.9 / 15.1 / 18.8	−6.5 / +14.3 / −1.6
Chen et al. (1988a,b), 2nd blocks	b	40 / 36	11.0 / 15.8	−6.4 / −10.0
Haseba (1973)	w / l	70 / 78	11.3 / 13.2	−10.6 / −5.4
Haseba and Ito (1984)	w / l	51 / 34	26.4 / 30.8	−5.7 / +2.4
Sparrow and Tien (1977)	w	34	7.1	+5.3
Sparrow et al. (1979)	w	33	22.0	−10.4
Parkhurst et al. (1968)	b	36	15.6	+9.5
Powell (1940)	w / l	12 / 52	32.6 / 35.3	+28.2 / −11.7
Schuepp (1993)	l	72	30.0	−1.9
Ramirez et al. (2002), thin subset	w / l	8 / 7	17.4 / 34.5	−14.4 / −5.3
Kobus and Shumway (2006)	b	17	36.3	+19.8

Table 10: Geometric mean of predicted/observed Nu by angle-of-attack bin.

β	0–15°	15–30°	30–45°	45–60°	60–75°	75–90°
windward	1.04	1.04	0.96	0.97	0.98	0.90
leeward	1.08	1.02	0.99	0.97	0.96	1.04
both	0.98	0.85	1.09	1.07	1.06	0.98

Data availability. The statistics in this section are those of the fit documented in `docs/validation.md`. The digitised measurement data set is not contained in the repository, so the validation cannot be re-run from the repository alone; the individual source tables in Table 9 account for 793 of the 815 fit rows.

10.3 Mixed convection

No measurements with simultaneous forced and free convection were available. The mixed model is therefore verified only in its limits ($u \to 0$: free convection; large u: forced convection, forced share $\to 1$). The data of Kobus and Wedekind (1996) (vertical plate, assisting and opposing flow, $Ri = 0.01$–1000) are a candidate for future validation.

11 Limitations and range of validity

- **Shape.** Flat, elliptical, undissected leaves with $AR \approx 1$–4; leaf curvature, lobes, serration, petioles and hairs are not represented. Edges carry no transfer.
- **Free convection.** Laminar core validated for leaf-sized ellipses and Ra_L typical of leaves; the turbulent branch is an extension with a generic coefficient (0.14) and has not been calibrated for leaves.

- **Forced convection.** Angle-of-attack model fitted for thin plates, $Re = 1.3\times10^3$–1.6×10^5, $\beta = 0$–$90°$, TI $\leq 22\,\%$. Extrapolation to canopy turbulence (TI $> 30\,\%$), to fluttering leaves and to leaves sheltered by neighbours is untested. Blunt bodies ($Re\,t/L \geq 2000$) are outside the fit.
- **Wind geometry.** The horizontal wind is assumed; vertical wind components and the reduction of the in-plane velocity component are not applied. Yaw (flow direction within the leaf plane) enters only through L_f.
- **Mixed convection.** Cubic blending without distinction of assisting and opposing flow; not validated against mixed-convection data.
- **Scalars.** The exponent k is an interpolation between modes; stomatal patchiness and the distribution of sources on the leaf surface are not considered.
- **Air properties.** Dry air with constant Pr and c_p; humidity enters only through Eq. (17).

12 Reproducibility

```
pip install .[test]                        # Python package graefe_convection
python -m pytest python -q                 # 17 Python tests
matlab -batch "cd matlab; runTests"        # 17 MATLAB tests
python python/make_reference.py            # regenerate reference values
python docs/manual/make_figures.py         # figures and numbers of this document
matlab -batch "cd docs/manual; ...
    matlab_crosscheck('generated/matlab_crosscheck.json')"
```

All figures and numerical values in Sections 3–9 are generated by `docs/manual/make_figures.py` from the model code and inserted into this document automatically. The software is licensed under GPL-3.0; please cite Graefe et al. (2022) when using the model (`CITATION.cff`).

A Source register

Table 11 lists every source cited, its role in the model or the validation and the verification status of its bibliographic data. Journal entries were verified against Crossref on 12 October 2026 (`docs/references.md`); the four additional sources used only in this document (Campbell and Norman, 1998; Monteith, 1973; Pohlhausen, 1921; Sutherland, 1893) were verified on 13 October 2026.

Table 11: Sources, their use and verification status (V = DOI verified via Crossref; – = no DOI).

Source	Use	Status
Graefe et al. (2022)	free-convection core (Eqs. 1, 3a–3g, 4–11, 14, 17, 18; Table 3)	V
Lienhard (2020)	laminar–transitional–turbulent average Nusselt number; uniform-flux coefficient 0.680	V
Mayle (1991)	transition onset vs. turbulence intensity, $Re_{\theta,t} = 400\,\mathrm{TI}^{-5/8}$	V
Kottke et al. (1977)	blunt-leading-edge criterion $Re_t \geq 2000$	V
Massman (1998)	diffusivities of H_2O and CO_2 in air	V
Yovanovich (1998)	conduction limit Nu_c (G22 Eq. 3f)	– (book chapter)
Yovanovich and Jafarpur (1993)	conduction limit of plates used in the validation blending	– (proceedings)

Source	Use	Status
Pohlhausen (1921)	laminar flat-plate coefficient 0.664	V
Sutherland (1893)	viscosity of air	V
Campbell and Norman (1998)	alternative laminar coefficient 0.6	V
Monteith (1973)	simple empirical g_b (`monteith_gb`)	–
Hassani and Hollands (1989)	free convection of disks and squares (validation)	V
Kobus and Wedekind (2001, 2002)	free convection of horizontal and inclined disks (validation)	V
Kobus and Wedekind (1996)	mixed convection, candidate data (not used)	V
Chen et al. (1988a)	forced, parallel flow, TI 3–22 %, two-sided	V
Chen et al. (1988b)	forced, angle of attack 0–180°, windward/leeward/two-sided	V
Haseba (1973)	elliptical leaf model 50 × 100 mm, angle of attack	V
Haseba (1974)	background, leaf-like plate	V
Haseba and Ito (1984)	strips of aspect ratio 0.05–20, windward and leeward	V
Sparrow and Tien (1977)	square plate, attack 25–90°, yaw 0–45°	V
Sparrow et al. (1979)	rectangles of aspect ratio 0.4 and 2.5	V
Parkhurst et al. (1968)	leaf models (ellipses), two-sided	V
Powell (1940)	evaporation plates, angle of attack 0–180°	–
Schuepp (1993)	leeward transfer (electrochemical); streamline-length weighting	V
Ramirez et al. (2002)	strip 18 × 170 mm, angle of attack 0–180°	V
Kobus and Shumway (2006)	crossflow and impinging disks	V
Motwani et al. (1985)	thick rectangular plates (blunt, excluded from fit)	V
aus der Wiesche and Helcig (2016)	thick disks (blunt, excluded from fit)	V
aus der Wiesche and Helcig (2020)	background, windward stagnation regime	V
Manna (2019)	excluded from the validation set (numerical results)	–

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